

AI Use Impact Tracker

v2 — Weighted Impact Index

Data Preparation, Computation Rules, and Implementation Status

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1. Overview

The AI Use Impact Tracker measures how AI use frequency relates to self-reported work impact across countries, gender, and age groups. It is built on survey data from the Global Mind Project (GMP) by Sapien Labs.

The v2 dashboard introduces the **Weighted Impact Index (WII)** as the primary outcome metric, replacing the simpler Net Impact Index (NII) from v1. The WII assigns severity weights to each impact flag, recognizing that job loss (−1.0) is more consequential than mild quality improvement (+0.5).

2. Data Source

Source: Global Mind Project (GMP), Sapien Labs. Monthly CSV extract of survey responses.

Current dataset: 266,589 rows (filtered to respondents with AI-related data) covering June 2025 – April 2026.

Data cleaning: Standard GMP cleaning criteria applied. Rows without ai_freq data are excluded. Pre-2025 months with near-zero adoption rates (< 5%) are dropped as they predate the AI questions.

Note: Per Tara's spec, the final version should use only Google Display and Meta traffic data, excluding Google Search and down-weighting organic traffic to 10%. This filtering is not yet implemented.

3. Key Variables

3.1 Exposure: ai_freq

A 7-level ordinal scale measuring how often a respondent uses AI assistants:

- 0 = Never
- 1 = Rarely
- 2 = Monthly (a few times a month)
- 3 = Weekly (several days a week)
- 4 = Daily (several times a day)
- 5 = Constantly
- 6 = Always (all of the time)

3.2 Outcome: ai_impact_work

A multi-select, pipe-delimited field. Each respondent can select multiple impacts. The field is parsed into nine binary flags, each carrying a signed severity weight:

Impact Flag (ai_impact_work)	Weight	Category
Created new job or income opportunities	+1.00	Positive
Improved my work quality or output	+0.50	Positive
No impact	0.00	Neutral
Not sure	0.00	Neutral
Another impact not listed here	0.00	Neutral
Made me worry about the future of my job or industry	-0.25	Negative
Increased pressure to adapt or work faster	-0.50	Negative
Reduced my income or made it harder to find work	-0.75	Negative
Caused me to lose my job	-1.00	Negative

4. Weighted Impact Index (WII)

4.1 Per-Respondent Score

For each respondent in the impact denominator, the per-respondent score is the dot product of their binary flag vector against the weight vector:

$$s_i = \sum_{f \in \mathcal{F}} w_f \cdot \mathbb{1}[\text{flag}_{f,i}]$$

where w_f is the weight for flag f from the IMPACT_WEIGHTS table, $\mathbb{1}[\text{flag}_{f,i}]$ is the indicator function (1 if respondent i selected flag f , 0 otherwise), and $\mathcal{F}$ is the set of eight weighted impact flags.

Example — per-respondent scores:

Respondent	Flags selected	Score
Respondent A	Improved quality, Adaptation pressure	$(+0.50) + (-0.50) = 0.00$
Respondent B	New job opportunities	+1.00
Respondent C	Job loss	-1.00
Respondent D	Improved quality, New opportunities	$(+0.50) + (+1.00) = +1.50$

Respondent E	Job anxiety, Reduced income	(-0.25) + (-0.75) = -1.00
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4.2 Stratum-Level Index

The WII for a stratum (e.g., a country-month) is the arithmetic mean of per-respondent scores across the impact denominator:

$$WII = (1 / N^d) \cdot \sum_{i \in \mathcal{D}} s_i$$

Example: Using the five respondents above:

$$WII = (0.00 + 1.00 + (-1.00) + 1.50 + (-1.00)) / 5 = +0.10$$

In practice, with tens of thousands of respondents per month, global WII values land around +0.14 to +0.18 — net positive, meaning the weighted benefit of AI on work slightly outweighs the weighted harm.

4.3 Impact Denominator

The denominator includes only respondents who: (a) use AI at any frequency ($ai_freq \geq 1$), and (b) have at least one non-null impact response. Respondents who selected only N/A or left the field blank are excluded.

4.4 Confidence Intervals

Two different confidence interval methods are used depending on the metric type:

Wald confidence interval (WII). The Weighted Impact Index is a mean of continuous per-respondent scores (ranging roughly from -1 to +1.5). Its 95% CI is computed via the standard error of the mean (SEM):

$$CI = WII \pm 1.96 \times \sigma_s / \sqrt{N^d}$$

$\sigma_s = \text{std dev of per-respondent scores}$

where SD is the standard deviation of the per-respondent scores and N is the impact denominator size. The constant 1.96 is the z-score from the standard normal distribution corresponding to 95% confidence (2.5% in each tail). This is justified by the Central Limit Theorem: with tens of thousands of respondents per month, the sampling distribution of the mean is approximately normal regardless of the underlying score distribution.

Example: For March 2026 with WII = +0.147 and ~25,000 respondents:

$$CI = 0.147 \pm 1.96 \times \sigma_s / \sqrt{25,000} = [0.140, 0.153]$$

The interval is tight (± 0.007) because N is large. The entire interval is above zero, meaning the positive weighted impact is statistically significant.

Wilson score interval (share metrics). Share-based metrics — adoption rate, individual impact flag shares, positive/negative impact shares — are proportions bounded between 0 and

1. For these, we use the Wilson score 95% confidence interval, which performs better than the Wald interval when proportions are near 0 or 1 or when sample sizes are smaller. This is the same method used in the v1 dashboard.

Why two methods? The WII is a continuous mean, not a proportion, so the Wald/SEM approach is appropriate. The share metrics are bounded proportions where Wilson score intervals provide better coverage, especially at the extremes. Both use the 95% confidence level ($z = 1.96$).

4.5 Asymmetry by Design

The weight scheme is deliberately asymmetric: the most severe negative outcome (job loss, -1.0) carries the same absolute weight as the strongest positive (new opportunities, $+1.0$), but the cumulative negative weight across all four negative flags (-2.5) exceeds the cumulative positive weight ($+1.5$). This reflects the stakeholder judgment that **severe harms deserve disproportionate weight** compared to moderate benefits.

4.6 Comparison with v1 Net Impact Index

The v1 NII is a simpler binary measure: $NII = \text{positive_share} - \text{negative_share}$. A respondent counts as positive if they selected any positive flag and negative if they selected any negative flag. NII treats all impacts equally; WII differentiates by severity.

In practice, global WII values (~ 0.14 – 0.18) run lower than NII values (~ 0.20 – 0.26) because the negative weights are heavier.

4.7 Rolling-Window Pooling

When the dashboard is set to a rolling window (e.g., "Last 3 months"), it computes a weighted average of the pre-computed monthly metrics, using each month's respondent count as the weight:

$$WII_{\text{pooled}} = \sum_{m=1 \dots w} (N_m \cdot WII_m) / \sum N_m$$

m = 1 ... w (months in window)

Example — Last 3 months pooled WII:

Month	N respondents	WII	N × WII
Feb 2026	20,000	+0.161	3,220
Mar 2026	25,000	+0.147	3,675
Apr 2026	15,000	+0.141	2,115
Pooled	60,000	+0.150	9,010

$$WII_{\text{pooled}} = 9,010 / 60,000 = +0.150$$

March contributes more because it has the most respondents (25K). This is more accurate than a simple average of the three monthly values, which would treat all months equally regardless of sample size.

Approximation note: The impact metrics (WII, positive/negative shares) technically have a different denominator — the impact denominator (AI users who answered the impact question), not total N. We use total N as a proxy because the two track closely within a short window, keeping the approximation within ~1 percentage point of exact.

5. Suppression Rule

Strata with fewer than **N = 50** respondents are suppressed. This prevents noisy estimates from small cells.

6. Stratification

Metrics are computed at eight stratum levels:

- Global (all respondents in a month)
- Country
- Gender
- Age band (9 bands: 18–20 through 85+)
- Country × Gender
- Country × Age band
- Gender × Age band
- Country × Gender × Age band

7. v2 Dashboard Features

7.1 Overview Tab

- Hero KPI card: Weighted Impact Index with 95% CI
- Supporting KPIs: respondent count, adoption rate, impact denominator
- WII time-series (monthly, global) with selection indicator
- Adoption rate bar chart (monthly, global)
- Impact composition chart with flag weights annotated + survey wording toggle
- WII by gender and age band (horizontal bar charts)

7.2 Countries Tab

- Top 12 countries ranked by WII
- Adoption rate × WII scatter plot with min-N slider (50–500)
- Full country-month table, sortable by any column

7.3 Frequency & Impact Tab

- Dose-response: NII by AI frequency level (global, by gender, by age)
- Frequency distribution stacked bar chart over time

7.4 Interactive Controls

- Month selector with prev/next navigation
- Rolling-window options: Last 3 months, Last 6 months (pooled)
- Min-N slider on scatter chart
- Short labels / survey wording toggle on composition chart

8. Technical Architecture

8.1 Pipeline

- CSV ingest → normalization (normalize.py) → metric aggregation (metrics.py) → Parquet output → HTML dashboard bake
- Pure-function pipeline: normalization and metric layers have no I/O side effects
- ETL runs in ~17 seconds on 266K rows across 8 stratum levels

8.2 Deployment

- Flask web app on Railway with Docker (python:3.12-slim)
- GitHub auto-deploy on push to main
- CSV upload triggers threaded ETL + dual dashboard bake (v1 and v2)
- Routes: /dashboard-v1/<id>, /dashboard-v2/<id>, /latest-v1, /latest-v2

- Static HTML works offline from file:// (Observable Plot + d3 from CDN)

9. Implementation Status

Feature	Status
Weighted Impact Index (per-flag weights)	Implemented
KPI dashboard with hero WII card + 95% CI	Implemented
WII time-series (monthly, global)	Implemented
WII breakdowns by gender, age, country	Implemented
Impact composition chart with weight annotations	Implemented
Country ranking by WII (top 12 bar chart)	Implemented
Adoption x WII scatter plot with min-N slider	Implemented
Sortable country table with WII column	Implemented
Dose-response (NII by frequency level)	Implemented
Rolling-window views (3-month, 6-month pooling)	Implemented
Separate v1/v2 dashboard routes on Railway	Implemented
Demographic-weighted averaging (UN age distributions)	Planned
Quarterly aggregation	Planned
Regional groupings and averages	Planned
Map views (choropleth)	Planned
Data source filtering (Google Display, Meta only)	Planned

10. Limitations

10.1 No Survey Weights (Non-Probability Sample)

The GMP respondent pool is a convenience sample recruited via online ads (Google Display, Meta, and organic search). It is not a probability sample and carries no post-stratification or design weights. Consequently:

- Demographic proportions in the data do not match national population proportions. Countries, age groups, and genders are represented according to who encounters and completes the survey, not according to census benchmarks.
- All metrics — WII, adoption rate, impact shares — are descriptive statistics of the responding population, not population-level estimates. We cannot claim, for example, that 42% of adults in a country use AI; we can only say that 42% of GMP respondents in that country reported using AI.
- Associations between AI frequency and impact outcomes are correlational. Higher-frequency AI users who report better outcomes may differ systematically from lower-

frequency users in education, occupation, digital literacy, or other unmeasured confounders. No causal claims are supported.

Planned mitigation: *Demographic-weighted averaging using UN World Population Prospects age–sex distributions is on the roadmap. This would rebalance age and gender composition within each country to approximate national demographics, reducing one source of composition bias. It would not, however, address self-selection into the survey or unmeasured confounders.*

10.2 Composition Effects in Time-Series

Month-over-month changes in WII or other metrics may reflect shifts in who is responding rather than genuine attitudinal change. For example:

- If a recruitment campaign increases responses from a younger demographic in one month, and younger respondents tend to report more positive AI impacts, the global WII will rise — even if no individual changed their view.
- Country mix changes: if a high-WII country (e.g., one with strong AI adoption) is over-represented in one month relative to another, the global metric shifts accordingly.
- Seasonal patterns in survey completion (e.g., lower response rates during holidays) can alter the demographic mix.

The stratified breakdowns (by country, gender, age) partially mitigate this by holding one dimension constant, but cross-dimensional composition effects remain. The rolling-window views help smooth short-term fluctuations.

10.3 Weight Selection

The nine impact weights (+1.0, +0.5, 0, −0.25, −0.50, −0.75, −1.0) were chosen by stakeholder judgment, not derived from empirical data. Different weight schemes would produce different index values and could alter relative country rankings. Sensitivity analysis (varying weights within a plausible range) has not yet been conducted. The asymmetric design — total negative weight (−2.5) exceeding total positive weight (+1.5) — is intentional but should be disclosed when comparing WII values across studies.

10.4 Rolling-Window Approximation

Rolling-window pooling (3-month, 6-month) computes a weighted average of pre-aggregated monthly metrics using each month's total respondent count as the weight. The exact calculation would re-aggregate from individual-level data within the pooled window. The approximation introduces minor error (± 1 percentage point within 3–6 month windows) because the impact denominator differs slightly from total N. This trade-off is accepted for client-side performance: the dashboard runs entirely in the browser and cannot re-query raw data.

10.5 Dose-Response Uses NII, Not WII

The Frequency & Impact tab shows dose-response curves (impact by AI frequency level) using the simpler Net Impact Index, not the Weighted Impact Index. This is because the current ETL computes NII stratified by frequency level but does not yet compute WII per frequency level. Adding weighted dose-response is a planned enhancement.

10.6 Data Source Filtering Not Yet Implemented

Per Tara's specification, the production version should restrict the respondent pool to Google Display and Meta traffic sources and down-weight organic/search traffic to 10%. This filtering is not yet implemented in the ETL pipeline. The current dashboard processes *all* traffic sources equally, which may include respondents with different characteristics (e.g., Google Search respondents may have higher digital literacy). Once implemented, historical metrics may shift as the composition of the respondent pool changes.

10.7 Multi-Select Interaction Effects

Respondents can select multiple impact flags simultaneously. The WII treats each flag independently — a respondent selecting both “improved quality” (+0.5) and “job anxiety” (−0.25) receives the sum (+0.25). This additive model does not capture potential interactions: for instance, the experience of simultaneous benefit and anxiety may not be well-described by simply adding the weights. More complex models (e.g., interaction terms, latent factor analysis) could address this but are beyond the current scope.

10.8 Suppression and Small-Cell Noise

Strata with fewer than 50 respondents are suppressed entirely. This threshold is a pragmatic choice; it prevents the noisiest estimates from appearing but does not guarantee statistical reliability. For strata just above the threshold (e.g., $N = 55$), confidence intervals are wide and point estimates can shift substantially with small data changes. Users should exercise caution when interpreting metrics from low- N strata, even when displayed.

10.9 Confidence Intervals Assume Simple Random Sampling

All confidence intervals — both Wald (for WII and frequency mean) and Wilson score (for share metrics) — assume simple random sampling. The GMP respondent pool is a convenience sample, so these CIs may understate the true uncertainty. They will be revised once survey weights are introduced.

11. Access

Web app: <https://ai-use-impact-tracker-demo.up.railway.app/>

v1 dashboard: /latest-v1 (Net Impact Index)

v2 dashboard: /latest-v2 (Weighted Impact Index)

GitHub: <https://github.com/jerzybala/ai-use-impact-tracker>